

Project: Applications of Electronic Circuits in Communication Engineering

I. Context & Problem

In wireless communication, the signal quality can be reduced by obstacles, fading, and propagation loss. A promising solution is the **Intelligent Reflecting Surface (IRS)**, which is made of many small electronic elements that can control how electromagnetic waves are reflected. From an electronics viewpoint, the behavior of each IRS element depends on practical circuit properties such as resistance, capacitance, and inductance.

In this project, students study a communication system assisted by IRS and focus on the problem of **maximizing the achievable rate** by optimizing the transmitter beamforming and the phase shifts of the IRS elements. Students will read the paper, understand the system model, implement one optimization algorithm, and compare their results with those reported in the paper

Reference: Class Materials

II. Algorithms:

1. Particle Swarm Optimization (PSO)
2. Ant Colony Optimization (ACO)
3. Artificial Bee Colony (ABC)
4. Memetic Algorithm (MA)
5. Whale Optimization Algorithm
6. Grey Wolf Optimization (GWO)
7. Firefly Algorithm (FA)
8. Salp Swarm Algorithm (SSP)
9. Genetic Algorithm (GA)
10. Evolution Strategy (ES)
11. Differential Evolution (DE)

III. Requirements:

- Each student chooses one algorithm from Section II and writes a report including: the basic description of the algorithm, pseudocode, computational complexity, convergence behavior, and possible variants (if any).
- Students must read and understand the paper, then write a summary report on the communication system and present the problem formulation in the paper.
- Students must reproduce the results reported in the paper mentioned in Section I. The report should clearly explain how the selected algorithm in Section II is applied and compare the implementation results with those in the original paper.